

SLE

- The hallmark of lupus is autoantibody production against many self antigens
- **Female** predominance varies from **4 : 1** before puberty to **8 : 1** afterwards

TABLE 157-1 -- Presenting Manifestations of Systemic Lupus Erythematosus

TARGET ORGAN	MANIFESTATIONS
Constitutional	Fatigue, anorexia, weight loss, prolonged fever, lymphadenopathy
Musculoskeletal	Arthralgias, arthritis
Skin	Malar rash, discoid lesions, livedo reticularis, vasculitis
Renal	Glomerulonephritis, hypertension, nephrotic syndrome, renal failure
Cardiovascular	Pericarditis (cardiac tamponade)
Neurologic	Seizures, psychosis, stroke, cerebral venous thrombosis, pseudotumor cerebri, aseptic meningitis, chorea, global cognitive deficits, mood disorders, transverse myelitis, peripheral neuritis (mononeuritis multiplex)
Pulmonary	Pleuritic pain, pulmonary hemorrhage
Hematologic	Coombs-positive hemolytic anemia, anemia of chronic disease, thrombocytopenia, leukopenia
Laboratory	Elevated ESR and CRP; decreased C3 or C4; positive ANA and anti-double-stranded DNA antibodies

TABLE 157-2 -- 1997 Revised Classification Criteria for Systemic Lupus Erythematosus

CRITERION ^[1]	DEFINITION
Malar rash	Fixed erythema, flat or raised, over the malar eminences, tending to spare the nasolabial folds
Discoid rash	Erythematous raised patches with adherent keratotic scaling and follicular plugging; atrophic scarring may occur in older lesions
Photosensitivity	Rash as a result of unusual reaction to sunlight (elicited by patient history or physician observation)
Oral ulcers	Oral or nasopharyngeal ulceration, usually painless, observed by a physician
Arthritis	Non-erosive arthritis involving two or more peripheral joints, characterized by tenderness, swelling, or effusion
Serositis	Pleuritis:convincing history of pleuritic pain or rub heard by a physician or evidence of pleural effusion
	or
	Pericarditis:documented by ECG or rub or evidence of pericardial effusion
Renal disorder	Persistent proteinuria >0.5 g/day or >3-plus (+ + +) if quantitation not performed
	or
	Cellular casts: may be red blood cell, hemoglobin, granular, tubular, or mixed
Neurologic disorder	Seizures:in the absence of offending drugs or known metabolic derangements (e.g., uremia, ketoacidosis, or electrolyte imbalance)
	or
	Psychosis:in the absence of offending drugs or known metabolic derangements (e.g., uremia, ketoacidosis, or electrolyte imbalance)
Hematologic disorder	Hemolytic anemia, with reticulocytosis
	or
	Leukopenia: <4,000/mm ³ total on two or more occasions
	or
	Lymphopenia: <1,500/mm ³ on two or more occasions
	or
	or
	Thrombocytopenia: <100,000/mm ³
Immunologic disorder	Anti-DNA antibody to native DNA in abnormal titer
	or
	Anti-Smith:presence of antibody to Smith nuclear antigen
	or
	Positive finding of antiphospholipid antibodies based on (1) an abnormal serum level of IgG or IgM anticardiolipin antibodies;(2) a positive test result for lupus anticoagulant using a standard method, or (3) a false-positive serologic test for syphilis known to be positive for at least 6 mo and confirmed by <i>Treponema pallidum</i> immobilization or fluorescent treponemal antibody absorption test (FTA-ABS). Standard methods should be used in testing for the presence of antiphospholipid.

- Drug-induced lupus is a lupus-like disease that is precipitated by exposure to certain drugs, notably many anticonvulsants, sulfonamides, and antiarrhythmic agents
- **Ixn** - ANA, Anti dsDNA, Anti Smith
serum complement

Mgt

- ✓ **Non steroidal drugs:** Arthritis and arthralgia, if not nephritis.
- ✓ **Anticoagulants:** in patients with thrombosis and anti-phospholipid antibodies.
- ✓ **Corticosteroids:** mainstay of therapy.
- ✓ **Cyclophosphamide.**
- ✓ Five year survival is now **>90%**.
- ✓ **Death:** is usually because of **nephritis**, **CNS** complications, and **Infections, pulmonary hemorrhage or myocardial infarction**

JRA

Three principal types of onset:

- ✓ Oligoarthritis or pauciarticular disease: Predominantly affects the joints of the lower extremities, only a single joint is involved at onset.
- ✓ Polyarthritis : Inflammation of only ≥ 5 joints is required as a criterion for classification of this type of onset
- ✓ Systemic-onset disease : prominent visceral involvement that includes hepatosplenomegaly, lymphadenopathy, and serositis. It is characterized by a quotidian(every day) fever with temperatures to $\geq 39^{\circ}\text{C}$, sometimes followed by mildly hypothermic temperatures for ≥ 2 wk. Each febrile episode is frequently accompanied by a characteristic faint, erythematous, macular rash; These evanescent salmon-colored lesions may be linear or circular, from **2–5** mm in size, and are often distributed in groups with a linear distribution most commonly over the trunk and proximal extremities
- The Koebner phenomenon, which is cutaneous hypersensitivity to superficial trauma resulting in a localized recurrence of the rash, is suggestive, but not diagnostic, of systemic-onset disease. Heat, such as a warm bath, also evokes a reappearance of the rash

TABLE 154-1 -- Criteria for the Classification of Juvenile Rheumatoid Arthritis

Age at onset: <16 yr
Arthritis (swelling or effusion, or the presence of 2 or more of the following signs: limitation of range of motion, tenderness or pain on motion, increased heat) in ≥ 1 joints
Duration of disease: ≥ 6 wk
Onset type defined by type of articular involvement in the 1st 6 mo after onset:
Polyarthritis: ≥ 5 inflamed joints
Oligoarthritis: ≤ 4 inflamed joints
Systemic disease: arthritis with a characteristic intermittent fever
Exclusion of other forms of juvenile arthritis

- C/F – Initial symptoms may be subtle or acute, and often include **morning stiffness** and gelling, **easy fatigability**, particularly after school in the early afternoon, **joint pain** later in the day, and objective **joint swelling**. The involved joints are often **warm**, resist full **range of motion**, are painful on motion, but are **not usually erythematous**

- **Ixn** – ANA, CBC(anemia and thrombocytopenia), RF, ESR, CRP,
X-ray : soft tissue swelling, regional osteoporosis, and periosteal new-bone apposition about affected joints
- **Rx**
Goal: preserve joint **function** and approximate **normal life style** with **few side effects** possible.
 - NONSTEROIDALS: including aspirin, are the first line drugs.
 - METHOTREXATE: safest second line drug (replaced gold salts).
 - ANTIMALARIALS: occasionally used.
 - CORTICOSTEROIDS: few indications.
 - PHYSIOTHERAPY AND OPHTHALMOLOGIC FOLLOW UP: extremely important.